

CSE110 Fall 2025 Lab Exam - 01 Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;

public class Flight007{
    public static void main (String[] args){
        Scanner sc = new Scanner(System.in);
        double luggage = sc.nextDouble();
        int seat = sc.nextInt();
        int counter = 4;
        double extra = 0;

        if(luggage>35){
            extra = (luggage - 35)* 3 + seat * 21;

            if (extra>300){
                counter = 1;
            }
            else if (extra>200 && extra<=300){
                counter = 2;
            }
            else if (extra>100 && extra<=200){
                counter = 3;
            }
        }
        System.out.println("You have to pay BDT "+extra);
        System.out.println("Please go to counter "+counter);
    }
}
```

RUBRIC

SL	Points To Meet	Marks [10]
<u>1</u>	Imports Scanner	0.5
<u>2</u>	Takes two inputs (Integer or double)	1
<u>3</u>	Initiates variables counter and extra before if/else	2
<u>4</u>	Checks if Luggage weight is over 35	1
<u>5</u>	Calculated the extra charge before the nested if/else	1.5
<u>6</u>	Checks extra charge conditions	2
<u>7</u>	Stores the counter number	1
<u>8</u>	Prints the two statements	1 (0.5+0.5)

[SET-B]

SOLUTION

```
import java.util.Scanner;
public class zuraClub{
    public static void main(String [] args){
        Scanner sc = new Scanner(System.in);
        int age = sc.nextInt();
        int years = sc.nextInt();
        double discount;
        double fee = 5000;
        if(years<2){
            discount = 0;
        }
        else{
            if(age<18 || age>40){
                discount = 0.10;
            }
            else if(age>=18 && age<=40 && years>=5){
                discount = 0.25;
            }
            else{
                discount = 0.15;
            }
        }
        double discounted = fee*discount;
        fee = fee - discounted;
        System.out.println("You get a discount of BDT "+discounted);
        System.out.println("You have to pay BDT "+fee);
    }
}
```

RUBRIC

SL	Points To Meet	Marks [10]
<u>1</u>	Imports Scanner	0.5
<u>2</u>	Takes two integer inputs	1
<u>3</u>	Initiates variable discount outside if/else	1
<u>4</u>	Uses the fixed value of fees (5000)	0.5
<u>5</u>	Checks if membership years is greater than 2	1
<u>6</u>	Checks age and years range	2
<u>7</u>	Assigns the discount percentage properly	1
<u>8</u>	Calculates the discount	1
<u>9</u>	Calculates the fee to pay	1
<u>10</u>	Prints the print statements	1 (0.5+0.5)

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Question 1 [C01] [10 Points]

Flight 007 has a luggage allocation of 35kg per customer. For any **extra luggage** the customer has to pay an **extra charge** at a specified counter. The extra charge is calculated using the formula, **Extra Charge = (Luggage Weight - 35) * 3 + Seat Number * 21**

You are tasked to take the luggage weight and seat number from the customer as input and calculate the extra charge and print which counter they have to go to based on the given table.

Conditions	Counter Number
Extra Charge <= 100	4
100 < Extra Charge <= 200	3
200 < Extra Charge <= 300	2
Extra Charge > 300	1

Sample Input	Sample Output
48 6	You have to pay BDT 243.0 Please go to counter 2
33 56	You have to pay BDT 0.0 Please go to counter 4

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Question 1 [C01] [10 Points]

“Zura” club has a membership renewal fee of BDT 5000. However, based on the age and the years of membership the club offers a discount on the renewal fee. The conditions for the discounts are as follows:

1. For members below the age of 18 and over the age of 40, they offer a discount of 10%.
2. For members aged between 18 and 40 with at least 5 years of membership the discount is 25% while those with less than 5 years of membership get the discount of 15%.
3. For members with less than 2 years of membership, there is no discount available.

You are tasked to **calculate the discount** and print the **amount of renewal fee** they have to pay to the gym. The members will provide you with their age and years of membership as input.

Sample Input	Sample Output
34 6	You get a discount of BDT 1250.0 You have to pay BDT 3750.0
42 1	You get a discount of BDT 0.0 You have to pay BDT 5000.0

CSE110 Fall 2025 Lab Exam - 02 Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;
public class TemperatureScore {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n;
        System.out.print("Enter n: ");
        n = sc.nextInt();
        int temperatureScore = 0;
        for (int i = 1; i <= n; i++) {
            int temp;
            System.out.print("Enter temperature " + i + ": ");
            temp = sc.nextInt();

            if (temp < 50 || temp > 150) {
                System.out.println("Invalid temperature, "+temp+".");
                i -= 1; // re-prompt for same reading
            } else {
                if (temp % 5 == 0 || temp % 4 == 0) {
                    temperatureScore += temp;
                } else {
                    temperatureScore -= temp;
                }
            }
        }
        int tempScoreCopy = temperatureScore;
        int evenDigitCount = 0;
        if (tempScoreCopy < 0) {
            tempScoreCopy *= -1;
        }
        while (tempScoreCopy != 0) {
            int digit = (tempScoreCopy % 10);
            if (digit % 2 == 0) {
                evenDigitCount++;
            }
            tempScoreCopy /= 10;
        }
        System.out.println("\nFinal Temperature Score: " + temperatureScore);
        System.out.println("Number of even digits: " + evenDigitCount);
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
<u>1</u>	Imports Scanner & takes input n & temperature	1.5
<u>2</u>	Runs the first loop n times.	1
<u>3</u>	Checks whether the temperature is within the range 50–150 and reprompts if it is not.	1.5
<u>4</u>	Initializes the variable temperatureScore and updates it according to the correct conditions.	2.5
<u>5</u>	Count the number of even digits of the temperature score & check for negative numbers.	2.5
<u>6</u>	Prints the Temperature Score and digit count.	1

[SET-B]

SOLUTION

```
import java.util.Scanner;
public class RainfallScore {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n;
        System.out.print("Enter K: ");
        int K = sc.nextInt();
        int rainfallScore = 0;
        for (int i = 1; i <= K; i++) {
            int rainfall;
            System.out.print("Enter rainfall " + i + ": ");
            rainfall = sc.nextInt();

            if (rainfall < 0 || rainfall > 200) {
                System.out.println("Invalid rainfall value, "+rainfall+".");
                i--; // re-prompt for same reading
            } else {
                if (rainfall % 7 == 0 || rainfall%3==0) {
                    rainfallScore += rainfall;
                } else {
                    rainfallScore -= rainfall;
                }
            }
        }
        int scoreCopy = rainfallScore;
        int oddDigitCount = 0;
        if (scoreCopy<0){
            scoreCopy*=-1;
        }
        while (scoreCopy != 0) {
            int digit = (scoreCopy % 10);
            if (digit % 2 != 0) {
                oddDigitCount++;
            }
            scoreCopy /= 10;
        }
        System.out.println("\nFinal Rainfall Score: " + rainfallScore);
        System.out.println("Number of odd digits: " + oddDigitCount);
    }
}
```


CSE110 Spring 2024 Lab Exam - 02 Tentative Solutions and Rubrics

RUBRIC

SL	Points To Meet	Marks [10]
<u>1</u>	Imports Scanner & takes input n & rainfall values	1.5
<u>2</u>	Runs the first loop n times.	1
<u>3</u>	Checks whether the rainfall is within the range 0–200 and reprompts if it is not.	1.5
<u>4</u>	Initializes the variable rainfallScore and updates it according to the correct conditions.	2.5
<u>5</u>	Count the number of odd digits of the rainfall score & check for negative numbers.	2.5
<u>6</u>	Prints the Rainfall Score and digit count.	1

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Question 1 [C01] [10 Points]

Take n valid temperature readings from the user. Temperatures below 50 or above 150 are invalid. For such user inputs display: "Invalid temperature, temperature value." Continue to take user inputs until you have 'n' valid inputs. Finally, calculate a Temperature Score based on the following rules:

- ❖ Add the temperature value if,
 - value is divisible by 5
 - value is a multiple of 4
- ❖ Otherwise, subtract the temperature value.

Count the number of even digits in the Final Temperature Score. [Before counting, check if the score is negative. If it is, multiply it by -1.]

Sample Input	Sample Output
Enter n: 5 Enter reading 1: 60 Enter reading 2: 55 Enter reading 3: 64 Enter reading 4: 10 Enter reading 4: 70 Enter reading 5: 51	Invalid temperature, 10. Final temperature Score: 198 Number of even digits: 1
Explanation: 6 temperature inputs taken as one was invalid (10). Temperature score = $60 + 55 + 64 + 70 - 51 = 198$ The even digit of the Temperature score is 8, so the number of even digits = 1.	

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Question 1 [C01] [10 Points]

Take K daily valid rainfall measurements (in millimeters). A rainfall measurement is **invalid** if the value is **below 0 or above 200**. For such inputs display: "Invalid rainfall value, rainfall measurement". Continue to take rainfall values from the user until you have K number of valid inputs. Finally, calculate a **Rainfall Score** using the following rules:

- ❖ **Add** the rainfall value if,
 - value divisible by 3
 - value is a multiple of 7
- ❖ **Otherwise, subtract** the rainfall value.

Count the number of odd digits in the Final Rainfall Score. [Before counting, check if the score is negative. If it is, multiply it by -1.]

Sample Input	Sample Output
Enter K: 5 Enter rainfall 1: 12 Enter rainfall 2: 5 Enter rainfall 3: 28 Enter rainfall 4: 25 Enter rainfall 5: 212 Enter rainfall 5: 49	Invalid rainfall value, 212. Final Rainfall Score: 59 Number of odd digits: 2
Explanation: 6 rainfall values taken as one was invalid (212). Rainfall score = $12 - 5 + 28 - 25 + 49 = 59$ The odd digit of the Temperature score is 5 & 9 so, the number of odd digits = 2.	

CSE110 Fall 2025 Lab Exam - 03 Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;

public class Exam3SetA{
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a headline: ");
        String sentence = scanner.nextLine();

        boolean isStartOfWord = true;

        for (int i = 0; i < sentence.length(); i++) {
            char c = sentence.charAt(i);

            if (c == ' ') {
                isStartOfWord = true;
            } else {
                if (isStartOfWord) {
                    if (c >= 'a' && c <= 'z') {
                        System.out.println("Formatting error
detected");
                        break;
                    }
                    isStartOfWord = false;
                }
            }
        }

        if (isStartOfWord==false){
            System.out.println("Valid headline");
        }
    }
}
```

RUBRIC

SL	Points To Meet	Marks [10]
1	Correctly takes the headline as input and follows the sample input	1
2	Proper use of loop to check each character in the headline	1
3	Correctly identifies the next word using spaces or boolean check	2
4	Implements correct logic to stop checking immediately when a word starts with lowercase	3
5	Correctly checks whether the first letter of each word is uppercase or lowercase.	2
6	Produces correct output for both valid and invalid sentences	1

[SET-B]

SOLUTION

```
import java.util.Scanner;

public class Exam3SetB{
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a sentence:");
        String sentence = scanner.nextLine();

        int currentWordLength = 0;

        for (int i = 0; i < sentence.length(); i++) {
            char c = sentence.charAt(i);

            if (c != ' ') {
                currentWordLength++;

                if (currentWordLength > 5) {
                    System.out.println("Found mistake in the
sentence");
                    break;
                }
            } else {
                currentWordLength = 0;
            }
        }

        if (currentWordLength<5){
            System.out.println("Correct sentence");
        }
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
1	Correctly takes the sentence as input and follows the sample input	1
2	Proper use of loop to check each character in the sentence	1
3	Correctly identifies the next word using spaces or boolean check	2
4	Implements correct logic to stop checking immediately when a word exceeds 5 characters	3
5	Correctly counts characters in each word	2
6	Produces correct output for both valid and invalid sentences	1

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Question 1 [C02] [10 Points]

You are tasked with creating a headline validator. Write a Java program that takes a sentence as input from the user. The program must check the words in order to detect if any word starts with a lowercase letter. Note: **Every word must start with an Uppercase letter!**

If a word starting with a lowercase letter is found, the program must stop checking immediately and print: Formatting error detected

If the program finishes checking the whole sentence without finding any lowercase starting letters, it must print: Valid Headline

You are not allowed to use any built-in functions except .length(), .charAt(), .codePointAt()

Sample Input	Sample Output
Enter a headline : The Quick Brown Fox	Valid Headline Explanation : Every word starts with a capital letter
Enter a headline : The Quick brown Fox	Formatting error detected Explanation : The word "brown" starts with a lowercase 'b'. The program detects this violation at the start of the third word and stops.
Enter a headline : java Is Fun	Formatting error detected Explanation : The first word "java" starts with a lowercase 'j'.

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Question 1 [C02] [10 Points]

You are tasked with creating a text validator. Write a Java program that takes a sentence as input from the user. The program must check the words to ensure that **no single word is longer than 5 characters**.

If a word with more than 5 characters is found, the program must stop checking immediately and print: Found mistake in the sentence

If the program finishes checking the whole sentence without finding any long words, it must print: Correct sentence

You are not allowed to use any built-in functions except .length(), .charAt(), .codePointAt()

Sample Input	Sample Output
Enter a sentence : Hi how are you	Correct Sentence Explanation : The longest word is “are” and “you” (3 characters). All are under the limit.
Enter a sentence : Hello Friend	Found mistake in the sentence Explanation : The word “friend” has 6 characters, which exceeds the limit of 5.
Enter a sentence : I am a Doctor	Found mistake in the sentence Explanation : The word “Doctor” has 6 characters.

CSE110 Fall 2025 Lab Exam - 04 Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;

public class LabExam4SetA{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] basket_capacity = {5, 6, 7, 2, 3};

        System.out.print("Enter the number of fruit types: ");
        int n = sc.nextInt();
        int[] fruits = new int[n];

        System.out.println("Enter the quantity of each fruit
type:");
        for (int i = 0; i < n; i++) {
            System.out.print("For Fruit type " + (i + 1) + " :
");
            fruits[i] = sc.nextInt();
        }

        System.out.println("Initial basket capacities:");
        for (int c = 0; c < basket_capacity.length; c++ ) {
            System.out.print(basket_capacity[c] + " ");
        }
        System.out.println();

        for (int i = 0; i < n; i++) {
            int fruit = fruits[i];
            boolean placed = false;

            for (int j = 0; j < basket_capacity.length; j++) {
                if (basket_capacity[j] >= fruit) {
                    basket_capacity[j] -= fruit;
                    placed = true;
                    break;
                }
            }

            if (!placed) {
                System.out.println("Fruit type " + (i + 1) + "
```

CSE110 Fall 2025 Lab Exam - 04 Tentative Solutions and Rubrics

```
with quantity " + fruit + " could not be placed.");
    }
}

System.out.println("Updated basket capacities:");
for (int c = 0; c < basket_capacity.length; c++) {
    System.out.print(basket_capacity[c] + " ");
}
}
```

RUBRIC

SL	Criteria	Marks
1	Correctly takes the number of fruit types as input, creates the array and reads the quantity of each fruit type into an array.	2
2	Prints the initial basket capacities correctly before allocation.	0.5
3	Accurately assigns each fruit type to the first basket with sufficient capacity; skips baskets that cannot accommodate the fruit.	3
4	Correctly reduces the basket capacity by the number of fruits allocated.	1.5
5	Identifies and displays fruit types that could not be placed, along with their quantities.	2
6	Prints the updated capacities of all baskets after allocation.	1

[SET-B]

SOLUTION

```
import java.util.Scanner;

public class LabExam4SetB{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of students present: ");
        int n = sc.nextInt();
        int[] roll = new int[n];
        System.out.println("Enter roll numbers:");
        for (int i = 0; i < n; i++) {
            System.out.print("Student " + (i + 1) + ": ");
            roll[i] = sc.nextInt();
        }
        System.out.println("\nOriginal attendance list:");
        for (int i = 0; i < n; i++) {
            System.out.print(roll[i] + " ");
        }
        System.out.println();
        int evenCount = 0, oddCount = 0;
        int highest = roll[0];
        int lowest = roll[0];
        for (int i = 0; i < n; i++) {
            if (roll[i] % 2 == 0)
                evenCount++;
            else
                oddCount++;
            if (roll[i] > highest)
                highest = roll[i];
            if (roll[i] < lowest)
                lowest = roll[i];
        }
        int[] reordered = new int[n];
        int index = 0;
        for (int i = 0; i < n; i++) {
            if (roll[i] % 2 == 0) {
                reordered[index++] = roll[i];
            }
        }
        for (int i = 0; i < n; i++) {
            if (roll[i] % 2 != 0) {
                reordered[index++] = roll[i];
            }
        }
    }
}
```

CSE110 Fall 2025 Lab Exam - 04 Tentative Solutions and Rubrics

```
    }
    System.out.println("\nReordered attendance list:");
    for (int i = 0; i < n; i++) {
        System.out.print(reordered[i] + " ");
    }
    System.out.println();
    System.out.println("\nNumber of even roll numbers: " +
evenCount);
    System.out.println("Number of odd roll numbers: " +
oddCount);
    System.out.println("\nHighest roll number: " + highest);
    System.out.println("Lowest roll number: " + lowest);
    }
}
```

RUBRIC

SL	Points To Meet	Marks [10]
1	The program correctly accepts the number of students and their roll numbers, storing them in an array.	1
2	Displays the original roll numbers in the order entered.	0.5
3	Correctly counts the number of even and odd roll numbers.	1.5
4	Finds and displays the highest and lowest roll numbers correctly.	2.5
5	Reorders the array so all even numbers appear first, then odd numbers, preserving original order.	4
6	Displays the reordered list correctly.	0.5

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Question 1 [C03] [10 Points]

A farmer is having difficulty organizing fruits into baskets, each with a fixed capacity. Write a Java program that should take the **number of fruit types** as input, store **their quantities** in an integer array and display the initial basket capacities. Your task is to **assign these fruits to the baskets** by following the fruit allocation rules.

Fruit Allocation Rules:

- A basket can only hold fruits if its capacity is **equal to or greater than** the number of fruits to be placed. Otherwise, the basket is skipped.
- After placing fruits into a basket, the basket's capacity is **reduced** by the number of fruits allocated.

Finally, the program should display **updated capacities** of the basket and the **fruit types that could not be placed** in any basket, along with their corresponding quantities.

Given basket capacities : `basket_capacity[] = {5, 6, 7, 2, 3}`

Sample Input	Sample Output
Enter the number of fruit types: 3 Enter the quantity of each fruit type: For Fruit type 1 : 6 For Fruit type 2 : 9 For Fruit type 3 : 2	Initial basket capacities: 5 6 7 2 3 Fruit type 2 with quantity 9 could not be placed Updated basket capacities: 3 0 7 2 3
Explanation : • Fruit type 1 (6 fruits) is placed in the second basket (capacity 6). After allocation, the basket capacity is reduced to 0. • Fruit type 2 (9 fruits) cannot be placed in any basket because no basket has enough capacity, so it remains unplaced. • Fruit type 3 (2 fruits) is placed in the first basket (capacity 5). After allocation, the basket capacity is reduced to 3.	

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Question 1 [C03] [10 Points]

A teacher records the attendance of students using their roll numbers, but due to manual entry, the roll numbers are not arranged in a proper order. Write a Java program that should accept the **number of students present**, store their roll numbers in an **integer array** and display the **original attendance list**. After that, the program must **rearrange the roll numbers** so that **even numbers appear first followed by the odd numbers** by maintaining the **original order**. Finally, the program should **display the reordered attendance list**, along with the **counts for even roll numbers and odd numbers**, the **highest and lowest roll numbers**.

Constraints:

- The reordered list must be out-of-place.
- You cannot use any built-in function other than `.length`.

Sample Input	Sample Output
Enter number of students present: 8 Enter roll numbers: Student 1: 15 Student 2: 8 Student 3: 21 Student 4: 4 Student 5: 10 Student 6: 7 Student 7: 18 Student 8: 3	Original attendance list: 15 8 21 4 10 7 18 3 Reordered attendance list: 8 4 10 18 15 21 7 3 Number of even roll numbers: 4 Number of odd roll numbers: 4 Highest roll number: 21 Lowest roll number: 3
Explanation : In the sample input, the teacher enters 8 students and their roll numbers: 15, 8, 21, 4, 10, 7, 18, and 3. The program first displays these as the original attendance list. It then rearranges the roll numbers by placing all even numbers first (8, 4, 10, 18) followed by all odd numbers (15, 21, 7, 3), while keeping their original order. The program counts 4 even and 4 odd roll numbers. Finally, it identifies 21 as the highest roll number and 3 as the lowest roll number.	